

Probabilistic reasoning over time.

Temporal model: It represents how the system evolves over time.

⇒ Inference in temporal model:

Inference - asking ques to the model.

Types of inference in temporal model.

⇒ Filtering

⇒ Smoothing

⇒ Prediction

⇒ Most likely sequence.

Filtering: It is used to find the probability of the current state given all evidences so far.

Problem: Given.

yesterday	Today Rain	Today No rain
Rain	0.1	0.3
No rain	0.3	0.7

Weather	Umbrella	No umbrella
No rain		
Rain	0.9	0.1

$$P(\text{Rain}) = 0.5 \quad P(\text{No rain}) = 0.5$$

Given that the person carry an umbrella today, what is the probability that it is raining today.

$$P(\text{Rain} / \text{Umbrella}) = \frac{P(\text{Umbrella} / \text{Rain}) \cdot P(\text{Rain})}{P(\text{Umbrella})}$$

$$= \frac{0.9 \times 0.5}{0.45} = 0.45$$

$$P(\text{No rain} / \text{Umbrella}) = P(\text{Umbrella} / \text{No rain}) \cdot P(\text{No rain})$$

$$= (0.2)(0.5)$$

$$= 0.10$$

$$\text{Normalization} = \frac{0.45}{0.45 + 0.10} = 0.8182$$

$$P(\text{Rain} / \text{Umbrella})$$

$$\frac{0.45 + 0.10}{0.45 + 0.10}$$

$$= 0.8182$$

$$P(\text{no Rain} / \text{Umbrella}) = \frac{0.10}{0.45 + 0.10} = 0.1818$$

Question:

It is used to find the probability of the future state using given current state evidence.

Eg: Given today's observation what is the probability that it will rain tomorrow.

$$P(R_{t+1} / \text{Umbrella}) = P(\text{Rain}_{t+1} / \text{Rain}_t) \cdot P(\text{Rain}_t)$$

$$+ P(\text{Rain}_{t+1} / \text{No Rain}_t) \cdot P(\text{No Rain}_t)$$

$$= 0.4 \cdot 0.8182 + 0.3 \cdot 0.1818$$

$$= 0.62728$$

something

Used to find the probability of the past state given observations before and after

$$P(X_k | e_1 = t)$$

Two probabilities

⇒ Forward probability

⇒ Backward probability

Question:

Given that the person did not carry the umbrella on day 1 and carry the umbrella on day 2, what is the probability that it is raining on day 1.

Step 1:

Finding the forward probability:

⇒ Find state of the previous state.

$$P(X_1 / e)$$

We are going to find the state of previous day.

t_1 : No umbrella.

$$P(\text{Rain} / \text{No umbrella}) = P(\text{No umbrella} / \text{Rain})$$

* $P(\text{Rain})$

$$= 0.1 \times 0.5$$

$$= 0.05$$

$$P(\text{No rain} / \text{No umbrella}) \cdot P(\text{No umbrella} / \text{No Rain})$$

* $P(\text{No rain})$

$$= 0.8 \times 0.5$$

$$= 0.40$$

Normaliz.

$$P(\text{Rain} / \text{No umbrella}) = \frac{0.05}{0.05 + 0.40}$$

$$= 0.111$$

$$P(\text{No rain/No umbrella}) = \frac{0.46}{0.05 + 0.46}$$

$$= 0.8589.$$

Step = 2

calculating the backward probability

consider as rain on Day 1

$$P(x_2/x_1) =$$

consider as rain (x1) Day 1

$$P(x_2/x_1) = P(\text{Rain}_2)$$

Step-3: combining forward and backward probability.

$$P(X_1/E_1, E_2) \propto P(X_1/E_1) * P(E_2/X_1)$$

$$\text{For rain} = 0.111 \times 0.69$$

$$= 0.07661$$

$$\text{For not rain} = 0.889 \times 0.41$$

$$= 0.36445$$

Normalise

$$\text{For rain} = \frac{0.07661}{0.07661 + 0.36445}$$

$$= 0.1738$$

$$\text{For not rain} = \frac{0.36445}{0.07661 + 0.36445}$$

$$= 0.8262$$

Probability $\rightarrow 0.1738 + 0.8262 = 100\%$,
check,

Most likely sequence,

Vital-B algorithm is used to calculate this sequence.